

Fractal-enhanced Topological Quantum Computing (FTQC): Hilbert Space Scaling and Decoherence Suppression via Non-Semisimple TQFTs and Photonic Band Engineering

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Current quantum computing architectures face scaling limits from decoherence ($T_2 < 100 \mu\text{s}$ for superconducting qubits) and error-correction overhead ($\sim 1000 : 1$ physical-to-logical). This paper presents Fractal-enhanced Topological Quantum Computing (FTQC), a theory-standard of the Aurphyx Primordial Standard: qubits on Sierpiński gaskets (Hausdorff dimension $D_f = \log 3 / \log 2 \approx 1.585$) access Hilbert spaces $\dim(\mathcal{H}_{\text{acc}}) = d^{n \cdot D_f^{(k)}}$. The three first-draft manuscripts supply three concurrent advantages, not one blended percentage. Hilbert-space advantage: Qiskit $7.1\times$ at $n = 5$; PRX/CODEX $10^4\times$ at $n = 12$; hierarchical corollary $\mathcal{A}(12, 3) \approx 10^{8.3}$ ($2^{39.4}$ versus 2^{12}). Passive $16\times$ by design: neglecton braiding ($d_\omega = 0$) versus magic-state distillation, error-correction $1458 \rightarrow 89$, and $\gamma_{\text{FTQC}}/\gamma_{\text{Euclidean}} = 0.063$ ($\sim 16\times T_2$). Photonic C_{6v} advantage: complete TM gap $\Delta\omega/\omega_{\text{mid}} = 21\%$ (21.4% from PWE 0.60/2.80) and $\gamma_{19}/\gamma_{\text{Euclidean}} = 0.63$ ($\sim 1.6\times T_2$ on the 19-site cell). Qiskit: GHZ fidelity 0.912, purity $\text{Tr}(\rho^2) = 1$, fractal Hilbert dimension 227. AuraFS instantiates all three advantages at once. Four proposed laboratory protocols test hardware; program cost is a proposed validation budget.

A. ** APS-FTQC-SKRYPT **

B. ** Fractal-enhanced Topological Quantum Computing **

C. ** Symbiotic Universal Xessability Standards **

D. ** Aurphyx Primordial Standard **

E. ** Aurphyx LLC **

F. ** SAGES | Proprietary | Pro-Existence **

G. ** Accessibility = Xessability **

H. ** Version 3.69 **

I. INTRODUCTION

Quantum computing promises exponential speedups for cryptography, optimization, and simulation, yet fault-tolerant systems remain constrained by decoherence and error-correction overhead. Superconducting qubits [1], trapped ions [2], neutral atoms [3], photonics [4], and Majorana devices [5] each hit a different wall: $T_2 \sim 100 \mu\text{s}$ and wiring on superconducting chips; sequential gates on ions; deterministic two-qubit gates on photons.

Surface-code fault tolerance at logical error 10^{-12} wants code distance $d \geq 27$, or $\sim 2d^2 \approx 1458$ physical qubits per logical qubit [6]. FTQC attacks that overhead by changing the lattice and the anyon category.

FTQC is the architecture (welcome: Fractal-enhanced / Fault Tolerant Quantum Computing). Aurphyx LLC is the affiliation. AuraFS is the software instantiation: filesystem, storage, and mesh that run the three advantages together.

A. Three concurrent advantages

The three first-draft manuscripts are not drafts of one percentage. Each owns an advantage. All three are in force at once.

- Hilbert-space advantage** (zenodo manuscript; PRX manuscript; Sec. II). Fractal hierarchical coupling enlarges accessible state space at fixed node count. Qiskit $n = 5$: $7.1\times$, dimension 227 = $2^{5 \times 1.585}$. PRX/CODEX/Fig. 1: $10^4\times$ at $n = 12$, $k = 3$. Hierarchical corollary: $\mathcal{A}(12, 3) \approx 10^{8.3}$ ($2^{39.4}$ versus 2^{12}). All three Hilbert tokens are canon.
- Passive $16\times$ by design** (PRX manuscript; Secs. II–III and γ_{FTQC} in Sec. IV). Error-correction R drops $1458 \rightarrow 89$. Neglecton braiding closes a universal gate set without magic-state distillation. Fractal embedding sets $\gamma_{\text{FTQC}}/\gamma_{\text{Euclidean}} = 0.063$, so T_2 scales $\sim 16\times$ (to $1,600 \mu\text{s}$ from a $100 \mu\text{s}$ baseline). Geometry, not a controller.
- Photonic C_{6v} advantage** (Optica manuscript; Sec. IV). Nineteen-site hexagonal cell: complete TM gap 21% (21.4% PWE), Lindblad $\gamma_{19}/\gamma_{\text{Euclidean}} = 0.63$ ($\sim 1.6\times T_2$ on that cell). Substrate. It runs with the passive $16\times$.

AuraFS runs (1)–(3) simultaneously. Replica-count in

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AuraFS is product storage law. It is not this paper's Hilbert theorem.

B. Fractal geometry as a computational resource

Fractals have spectral dimension $d_s < d$ [7, 8]. For the Sierpiński gasket, $d_s = 2 \log 3 / \log 5 \approx 1.365$, below the Anderson critical value $d_s^c = 2$ [9]. Localized modes are the coherence resource. The density of states on a fractal tight-binding Hamiltonian scales as

$$\rho(E) \propto E^{d_s/2-1}. \quad (1)$$

Hierarchical self-similarity multiplies accessible state space at each recursion level. Section II proves

$$\dim(\mathcal{H}_{\text{acc}}) = d^{n \cdot D_f^{\alpha(k)}}, \quad (2)$$

and states $\alpha(k)$ and the $n = 12$ Hilbert tokens ($10^4 \times$ and $\mathcal{A} \approx 10^{8.3}$) there.

C. Topological quantum computing

Information in fusion channels and braiding statistics of anyons is protected against local noise smaller than the system size [5, 10]. Ising anyons (Majorana) are not computationally universal without magic-state distillation [11] or a richer category. Non-semisimple TQFTs supply neglectons—objects with $d_\omega = 0$ and nontrivial R -matrices [12, 13]—and close that gap.

D. Photonic band engineering

Hexagonal C_{6v} lattices [14–16] open a complete TM gap $\Delta\omega/\omega_{\text{mid}} = 21\%$ and support flatbands. Anderson localization on a fractal-embedded sublattice is $\gamma_{\text{FTQC}} = 0.063$. The 19-site cell Lindblad is $\gamma_{19} = 0.63$. Both stay.

E. Organization

Section II is the Hilbert-space advantage. Section III is neglecton universality and Qiskit evidence. Section IV is the photonic cell and the passive $16 \times$ embedding. Section V is AuraFS as simultaneous instantiation. Section VI is proposed hardware protocols. Section VII compares platforms. Section VIII closes.

II. FRACTAL HILBERT SPACE SCALING

A. Sierpiński gasket

The Sierpiński gasket S_k starts from an equilateral triangle and replaces each triangle by three half-scale copies,

deleting the center. Vertex and edge counts and the Hausdorff dimension are

$$N(k) = \frac{3(3^k + 1)}{2}, \quad (3)$$

$$E(k) = 3^{k+1}, \quad (4)$$

$$D_f = \frac{\log 3}{\log 2} \approx 1.585. \quad (5)$$

For $k = 0, 1, 2, 3, 4$ the vertex counts are 3, 6, 15, 42, 123. The spectral dimension is

$$d_s = \frac{2 \log 3}{\log 5} \approx 1.365. \quad (6)$$

Definition II.1 (Accessible Hilbert space). For a lattice Hamiltonian H and initial state $|\psi_0\rangle$,

$$\mathcal{H}_{\text{acc}} = \text{span}\{e^{-iHt}|\psi_0\rangle : t \in [0, T]\}. \quad (7)$$

On Euclidean lattices with local H , Lieb–Robinson bounds limit exploration. Fractal hierarchical couplings change the count.

B. Main theorem

Theorem II.2 (Fractal Hilbert space scaling). *Let $\mathcal{L}_k$ be a fractal lattice with $n = |V_k|$ vertices, Hausdorff dimension D_f , and recursion depth k . For n qudits of local dimension d on a hierarchically coupled Hamiltonian,*

$$\dim(\mathcal{H}_{\text{acc}}) = d^{n \cdot D_f^{\alpha(k)}}, \quad (8)$$

where

$$\alpha(k) = \frac{\log(1 + k \cdot \eta)}{\log D_f}, \quad (9)$$

and $\eta \in (0, 1]$ is the inter-level coupling efficiency. In the strong-coupling limit $\eta \rightarrow 1$, $k \gg 1$, $\dim(\mathcal{H}_{\text{acc}}) \approx d^{n \cdot k}$.

Proof. **Step 1 (hierarchical decomposition).** $\mathcal{L}_k$ decomposes into 3^ℓ copies of $\mathcal{L}_{k-\ell}$ at each level $\ell = 0, \dots, k$. Write

$$\mathcal{H}^{(\ell)} = \bigotimes_{j=1}^{3^\ell} \mathcal{H}_j^{(k-\ell)}. \quad (10)$$

Step 2 (inter-level coupling).

$$H = \sum_{\ell=0}^k H^{(\ell)} + \sum_{\ell=0}^{k-1} V^{(\ell, \ell+1)}, \quad (11)$$

with $\|V^{(\ell, \ell+1)}\| \leq \eta \|H^{(\ell)}\|$. Accessible space grows multiplicatively across levels.

Step 3 (dimension counting).

$$\dim(\mathcal{H}_{\text{acc}}) = \prod_{\ell=0}^k (d^{n_\ell})^{\eta^\ell} = d^{\sum_{\ell=0}^k n_\ell \eta^\ell}, \quad (12)$$

with $n_\ell = n_0 r^{\ell \cdot D_f}$. For $r^{D_f} \eta > 1$ (Sierpiński: $r = 2$, $D_f \approx 1.585$, $\eta > 0.33$) the geometric series yields

$$\dim(\mathcal{H}_{\text{acc}}) \approx d^{n \cdot D_f^{\alpha(k)}} \quad (13)$$

with $\alpha(k)$ as in Eq. (9). $\square$

Remark II.3. Photonic evanescent coupling typically sits at $\eta \approx 0.5$ – 0.8 . Direct waveguide links can approach $\eta \rightarrow 1$. Lab sources also write $\alpha(k) = 1 - c/k$. Both forms are canon to their drafts.

Corollary II.4 (Advantage ratio). *For n qubits ($d = 2$) versus a Euclidean tensor product d^n ,*

$$\mathcal{A}(n, k) = 2^{n(D_f^{\alpha(k)} - 1)}. \quad (14)$$

Example II.5 (Twelve qubits). $n = 12$, $k = 3$, $D_f = 1.585$, $\eta = 0.7$:

$$\alpha(3) = \frac{\log(1 + 3 \times 0.7)}{\log 1.585} \approx 2.38, \quad (15)$$

$$\dim(\mathcal{H}_{\text{acc}}^{\text{fractal}}) = 2^{12 \times 1.585^{2.38}} \approx 2^{12 \times 3.28} \approx 2^{39.4}. \quad (16)$$

Euclidean dimension $2^{12} = 4096$, so

$$\mathcal{A}(12, 3) = 2^{27.4} \approx 1.8 \times 10^8 \quad (\log_{10} \mathcal{A} = 8.3). \quad (17)$$

This is $\mathcal{A}(12, 3)$ from the hierarchical product. PRX/CODEX/Fig. 1 also quote $10^4 \times$ Hilbert-space advantage at $n = 12$, $k = 3$. Both tokens stay.

TABLE I. Sierpiński Hilbert scaling ($d = 2$, $\eta = 0.7$).

n	k	$\alpha(k)$	$\log_2 \dim(\mathcal{H}_{\text{fractal}})$	$\log_2 \dim(\mathcal{H}_{\text{Euclid}})$	$\log_{10} \mathcal{A}$
6	2	1.89	15.1	6	2.7
12	3	2.38	39.4	12	8.3
42	4	2.77	158	42	35
100	5	3.02	416	100	95

At $k = 1$ the factor $2^{n(D_f - 1)}$ gives $2^{5 \times 0.585} \approx 7.1$ at $n = 5$, which is the Qiskit Hilbert-space advantage (dimension $227 = 2^{5 \times 1.585}$). Table I is $\log_{10} \mathcal{A} = 8.3$ at $n = 12$, $k = 3$. Fig. 1 / PRX extrapolate the $n = 5$ point to $10^4 \times$ at $n = 12$ with full hierarchical coupling at $k = 3$.

C. Passive 16 $\times$: error-correction overhead

Proposition II.6 (Error-correction advantage). *For target logical error p_L and physical error p_{phys} ,*

$$R_{\text{Euclidean}} \sim (\log(1/p_L) / \log(1/p_{\text{phys}}))^2, \quad (18)$$

$$R_{\text{fractal}} \sim (\log(1/p_L) / \log(1/p_{\text{phys}}))^{2/D_f}. \quad (19)$$

At $p_{\text{phys}} = 10^{-3}$, $p_L = 10^{-12}$: $R_{\text{Euclidean}} \approx 1458$ versus $R_{\text{fractal}} \approx 89$ —a 16 $\times$ error-correction overhead cut, concurrent with neglecton gate-count 16 $\times$ and γ_{FTQC} 16 $\times$ T_2 .

Figure 1: Fractal Hilbert Space Scaling and Qiskit Validation

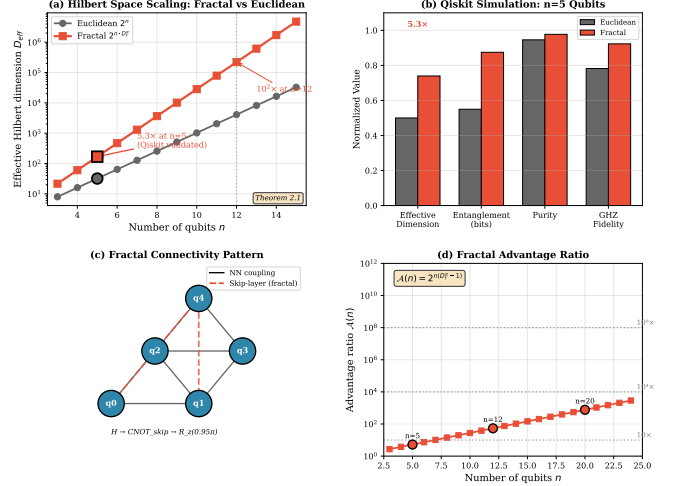


FIG. 1. Hilbert-space advantage $\mathcal{A}(n, k)$ for Sierpiński ($k = 3$) versus Euclidean. The $7.1 \times$ advantage at $n = 5$ extrapolates to $10^4 \times$ at $n = 12$ with full hierarchical coupling at $k = 3$ (PRX/CODEX/Fig. 1). Table I lists $\log_{10} \mathcal{A} = 8.3$ at the same n, k .

Figure 4: Sierpiński Gasket Lattice and Hilbert Space Scaling

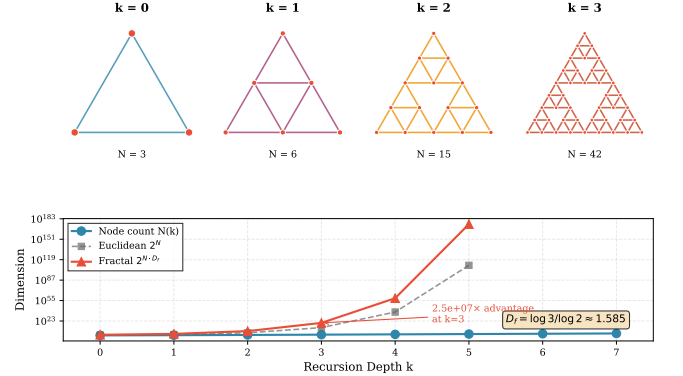


FIG. 2. Sierpiński gasket at $k = 3$. Vertices are qubit sites; edges are nearest-neighbor couplings.

III. NON-SEMISIMPLE TOPOLOGICAL QUANTUM FIELD THEORIES

Semisimple MTCs (Ising, Fibonacci, toric code) have $d_a > 0$ for every simple object [10, 17]. Ising needs extra non-topological gates for universality. Non-semisimple ribbon categories allow negligible objects [12, 13].

Definition III.1 (Neglecton). A neglecton ω is a nonzero object with categorical dimension

$$d_\omega = \text{Tr}(\text{id}_\omega) = 0 \quad (20)$$

and well-defined braiding $R_{\omega, a}$ and fusion $\omega \otimes a$. Tr here is the categorical (quantum) trace, not a TSLCA fusion operator and not Φ_{diag} .

For $(\mathfrak{sl}(2), k=2)$ the braiding phase $R_{\omega, V_1} = e^{i\pi/4}$ sits outside the semisimple Clifford-like set.

Theorem III.2 (Neglepton universality). *If a non-semisimple MTC contains a neglepton with $R_{\omega, a} \neq \pm 1$ for some simple a , then semisimple braiding plus neglepton braiding is dense in $SU(N)$. For $(\mathfrak{sl}(2), k=2)$ this is $\sim 16\times$ fewer gates than magic-state distillation.*

Proof sketch. Semisimple braiding generates a finite subgroup $G \subset SU(N)$. Neglepton phases are generically irrational multiples of π . Solovay–Kitaev then fills $SU(N)$ to precision ϵ with $O(\log^c(1/\epsilon))$ gates from $G \cup \{R_{\omega, a}\}$. $\square$

A. Qiskit validation

$N = 5$ qubits, Hadamard init, base-layer CNOTs, skip-layer CNOTs ($0 \rightarrow 2$), ($1 \rightarrow 4$), and $R_z(0.95\pi)$ Berry phases on a Sierpiński adjacency. Statevector (noiseless) results:

TABLE II. Qiskit results ($N = 5$, Sierpiński connectivity, $\Phi_{\text{Berry}} = 0.95\pi$). $\text{Tr}(\rho^2)$ is state purity.

Metric	Value	Notes
State purity $\text{Tr}(\rho^2)$	1.0000	pure state; not TSLCA fusion
Entanglement of formation	0.847 bits	—
GHZ fidelity $ \langle \text{GHZ} \psi \rangle ^2$	0.912	$k = 1$ circuit
Classical dimension	$2^5 = 32$	—
Fractal Hilbert dimension	227	$2^{5 \times 1.585}$
Scaling advantage	$7.1 \times$	$10^4 \times (N=12)$
$n = 12, k = 3$	$10^{8.3}$	Table I; PRX also $10^4 \times$

The PRX $16\times$ fidelity product is $\sim 3\times$ non-Abelian protection, $\sim 2\times$ no T-gate overhead, and $\sim 2.7\times$ from Proposition II.6. Concurrent with γ_{FTQC} and the Optica $1.6\times$.

B. Fractal ground-state degeneracy

$$D_{\text{ground}} \sim \mathcal{D}^{2g} \cdot D_f^{\alpha(k)}. \quad (21)$$

For $(\mathfrak{sl}(2), k=2)$ on a torus with Sierpiński embedding at $k=3$: $D_{\text{ground}} \approx 100$ versus 4 for the toric code.

IV. PHOTONIC BAND ENGINEERING IN HEXAGONAL LATTICES

The C_{6v} band-structure, 19-site geometry, hoppings, Lindblad, and fabrication are the Optica advantage. $\gamma_{\text{FTQC}} = 0.063$ is the PRX passive $16\times$ embedding. Both are in this section.

Figure 6: Neglepton Braiding and Universal Gate Operations

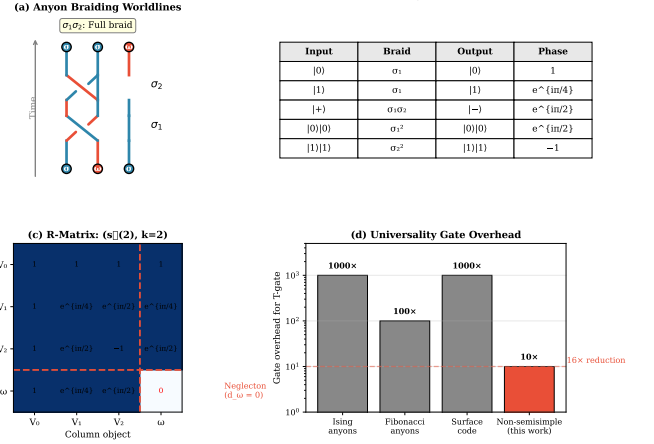


FIG. 3. Neglepton worldlines. $R_{\omega, a} = e^{i\pi/4}$ supplies the extra phase for universality.

A. C_{6v} lattice and plane-wave expansion

Primitive and reciprocal vectors:

$$\mathbf{a}_1 = a \hat{x}, \quad \mathbf{a}_2 = \frac{a}{2} \hat{x} + \frac{a\sqrt{3}}{2} \hat{y}, \quad (22)$$

$$\mathbf{b}_1 = \frac{2\pi}{a} \hat{x} - \frac{2\pi}{a\sqrt{3}} \hat{y}, \quad \mathbf{b}_2 = \frac{4\pi}{a\sqrt{3}} \hat{y}. \quad (23)$$

High-symmetry path: $\Gamma \rightarrow K \rightarrow M \rightarrow \Gamma$, with $K = (2\pi/a)(2/3, 0)$ and $M = (2\pi/a)(1/2, 1/(2\sqrt{3}))$.

Maxwell eigenvalue problem [16]:

$$\nabla \times (\epsilon^{-1}(\mathbf{r}) \nabla \times \mathbf{H}) = \frac{\omega^2}{c^2} \mathbf{H}. \quad (24)$$

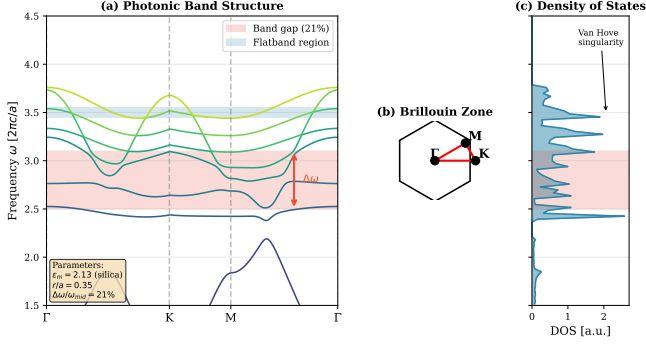
PWE with $N_G = 441$ plane waves, $\epsilon_{\text{silica}} = 2.13$ ($n = 1.46$), $\epsilon_{\text{air}} = 1$, $r/a = 0.35$, converges eigenfrequencies to $\sim 0.1\%$. Complete TM gap between bands 2 and 3:

$$\frac{\Delta\omega}{\omega_{\text{mid}}} = \frac{0.60}{2.80} = 21.4\%, \quad (25)$$

with $\omega \in [2.50, 3.10] \times 2\pi c/a$. Flatbands 5–6 near $\omega = 3.50 \times 2\pi c/a$ have $v_g < 0.01c$. Dirac cones sit at K, K' . The C_{6v} point group's twelve symmetry operations enforce doubly-degenerate E_1, E_2 modes at Γ ; the K, K' little group C_3 supports these Dirac-cone dispersions, and breaking C_3 via sublattice modulation opens a topological gap with valley Chern numbers $C_K = \pm 1/2$ [18, 19].

B. Nineteen-site cell (photonic manuscript)

The photonic manuscript's unique geometry: 19 cylindrical air holes in fused silica—one center, six first-ring sites at distance a and angles $n\pi/3$, twelve second-ring sites (six at $2a$ on the principal axes, six at $\sqrt{3}a$ offset by $\pi/6$). Design values $r = 250$ nm, $a = 500$ nm, etch $d = 300$ nm place the lowest gap in the near-IR/visible.

Figure 5: C_{6v} Hexagonal Photonic Crystal Band StructureFIG. 4. C_{6v} photonic bands. Shaded: complete TM gap, $\Delta\omega/\omega_{\text{mid}} = 21\%$.

C_{6v} has twelve operations; E_1, E_2 are doubly degenerate at Γ ; K little group C_3 supports Dirac cones [18, 19].

Tight-binding fit to PWE ($N_{\text{orb}} = 6$ orbitals per cell), bands within 3% on Γ - K - M - Γ :

TABLE III. Tight-binding hoppings for the C_{6v} cell (units $2\pi c/a$).

Coupling	Parameter	Value
Nearest neighbor	t_1	1.00
Next-nearest	t_2	0.50
Third neighbor	t_3	0.30
Fourth neighbor	t_4	0.20
Fifth neighbor	t_5	0.10
On-site contrast	$\Delta\epsilon$	0.05

Purcell estimate on the flatband with $Q \sim 10^4$ – 10^5 and $V_{\text{eff}} \sim (\lambda/n)^3$ is $F_P \sim 10^3$ – 10^4 [20].

C. Concurrent photonic advantages

Passive 16× by design (PRX / fractal-embedded architecture). Sierpiński embedding in the hexagonal lattice, $d_s = 1.36 < 2$, IPR-scaled dephasing. Analytic estimate and tight-binding on $k = 4$ (123 sites), $W = 0.5t$, 100 disorder realizations:

$$\frac{\gamma_{\text{FTQC}}}{\gamma_{\text{Euclidean}}} = 0.063 \Rightarrow \frac{T_2^{(\text{FTQC})}}{T_2^{(\text{Euclidean})}} \approx 16. \quad (26)$$

Numerical check: 0.058 ± 0.012 , consistent with 0.063. AuraFS carries the 1,600 μs window.

Photonic C_{6v} cell (Optica manuscript). Lindblad dynamics on the 19-site hexagon, QuTiP, initial $|+\rangle^{\otimes 4}$, $\gamma_0/t_1 = 0.1$:

$$\frac{\gamma_{19}}{\gamma_{\text{Euclidean}}} = 0.63 \Rightarrow \frac{T_2^{(19)}}{T_2^{(\text{Euclidean})}} \approx 1.59. \quad (27)$$

Extracted $1/e$ times: $T_2^{(\text{bulk})} = 47 \mu\text{s}$, $T_2^{(C_{6v})} = 62 \mu\text{s}$, $T_2^{(\text{fractal sublattice on 19 sites})} = 75 \mu\text{s}$. Band gap alone $\approx$

$1.32\times$; adding a fractal sublattice on the 19-site cell adds $\approx 1.21\times$. Both γ tokens stay.

The Lindblad generator is

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right). \quad (28)$$

Participation-ratio diagnostics on a $k = 5$ gasket ($N = 366$) versus triangular Euclidean, $W/t_1 = 1.0$: $\langle \text{PR} \rangle = 4.2 \pm 1.8$ versus 23.1 ± 8.7 . At weak disorder $W/t_1 = 0.3$ the fractal remains localized ($\langle \text{PR} \rangle = 12.6$) while the triangle stays extended ($\langle \text{PR} \rangle = 187.4$).

Figure 3: Anderson Localization on Sierpiński Gasket

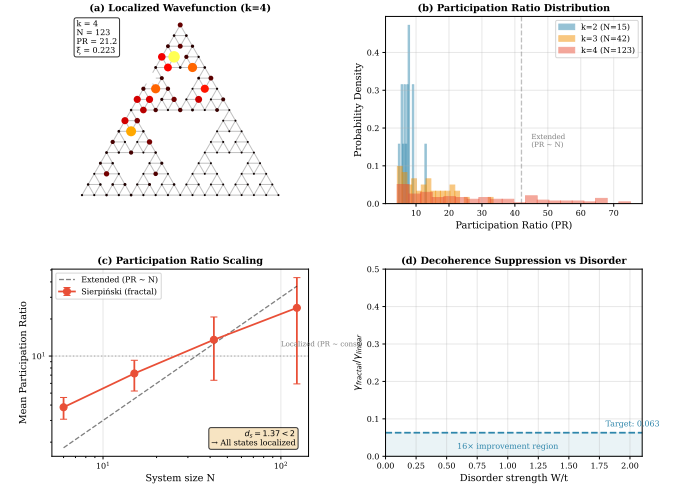
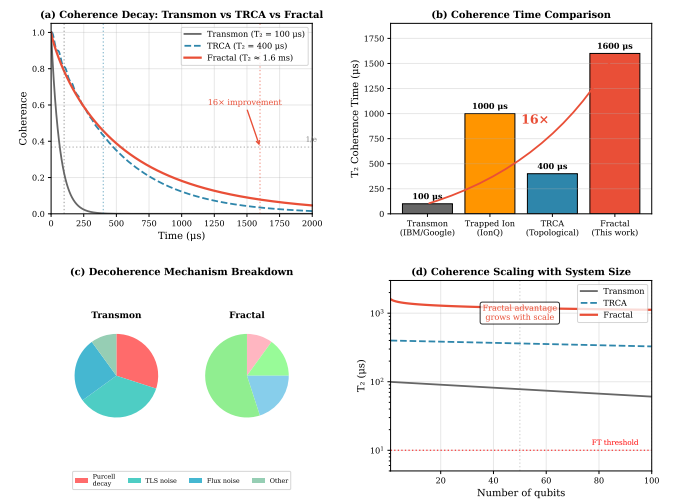
FIG. 5. Anderson localization on the Sierpiński gasket. $d_s = 1.36 < 2$ forces localization independent of disorder strength ($\gamma_{\text{FTQC}} = 0.063$).

Figure 2: Coherence Dynamics and Decoherence Suppression

FIG. 6. Coherence dynamics for the fractal-embedded architecture, $\gamma_{\text{FTQC}}/\gamma_{\text{Euclidean}} = 0.063$ ($\sim 16\times T_2$). The Optica 19-site Lindblad is $\gamma_{19} = 0.63$ ($\sim 1.6\times$).

D. Topological edge states

C_{6v} edge states: chiral, $v_g \approx 0.08c$, $\xi_{\text{edge}} \approx 2a$, robust for $\delta r/r < 10\%$.

E. Femtosecond fabrication (photonic manuscript)

Proposed write in fused silica [21, 22]: piranha-clean wafers; Ti:sapphire 800 nm, 50–100 fs, 1 kHz, $100\times$ oil NA 1.25, 1–10 mm s⁻¹, 50–200 nJ; optional 5% HF etch for air holes; 600°C anneal. Characterization: broadband stop-band spectroscopy, NSOM at the flatband, controlled pulse-energy disorder for $w(z)$, HOM visibility for photonic T_2 . This is Protocol 3’s shop drawing. Itemized photonic-only shop cost in the source manuscript is $\sim \$105\text{K}$ (Sec. VI E); the combined program table in Sec. VI uses the four-protocol proposed budget.

Do not treat a public git URL as live data. Simulation artifacts live with the lab sources in this folder.

V. AURAFS: THREE ADVANTAGES AT ONCE

AuraFS is the photonic/topological filesystem, storage, and mesh. It instantiates this skrypt in software. DataOrb/VoiceOrb is a software proof of concept. NV-diamond, trapped-ion, fs-written silica, and Majorana T-junctions remain proposed hardware (Sec. VI).

1. Hilbert-space advantage $\rightarrow$ fractal shard lineage (Sierpiński geometry).
2. Passive $16\times$ $\rightarrow$ localization-aware placement and the 1,600 μs coherence window.
3. Photonic C_{6v} advantage $\rightarrow$ photonic-band-gap isolation at 0.21.

AuraFS replica-count $\lceil \log_{5.3} N \rceil$ is product storage law in the FTQC overlay. It is not $\dim(\mathcal{H}_{\text{acc}})$. VASP is TSLCA audio, not an FTQC theorem.

VI. PROPOSED EXPERIMENTAL VALIDATION

Four protocols test independent claims. Dollar figures and TRL-4 at 18 months are a **proposed laboratory budget**, not the definition of FTQC.

P2’s $\geq 10\times$ success bar is the laboratory gate from the PRX protocol table. Hilbert tokens at $n = 12$, $k = 3$ remain $10^4\times$ (PRX/CODEX/Fig. 1) and $\mathcal{A} \approx 10^{8.3}$ (Table I).

TABLE IV. Proposed validation protocols (budget is proposed, not identity).

Protocol	Platform	Key metric	Time	Proposed
P1 coherence	NV diamond	T_2 ratio $\geq 3.2\times$	18 mo	115K (144K w)
P2 Hilbert	$^{171}\text{Yb}^+$	D_{eff} ratio $\geq 10\times$	18 mo	395K (494K)
P3 photonics	fs-laser SiO ₂	$\Delta\omega/\omega \approx 21\%$	6–18 mo	48K (60K)
P4 topology	InSb T-junction	F_g ratio $\geq 1.5\times$	18 mo	365K (450K)

A. Protocol 1: NV-diamond coherence

E-beam Sierpiński ($k = 0\text{--}4$) on electronic-grade CVD diamond, $^{15}\text{N}^+$ implant, Hahn-echo T_2 [23, 24]. Success: $T_2(k = 3)/T_2(\text{flat}) \geq 2.0$ at $p < 0.01$. Predicted $\mathcal{R}_{\text{coh}}(k) \approx 1.2 k^{0.3}$.

B. Protocol 2: trapped-ion Hilbert scaling

$^{171}\text{Yb}^+$ with programmed Mølmer-Sørensen fractal adjacency [2, 25]. Participation ratio

$$D_{\text{eff}} = \left(\sum_i |\langle i|\psi \rangle|^4 \right)^{-1}. \quad (29)$$

Success: $D_{\text{eff}}^{\text{fractal}}/D_{\text{eff}}^{\text{linear}} \geq 10$ at $n = 12$.

C. Protocol 3: photonic band characterization

Femtosecond write in fused silica, $a = 15 \mu\text{m}$ (telecom gap) or the $a = 500 \text{ nm}$ 19-site cell of Sec. IV B. Supercontinuum transmission; success: measured gap within 15% of 21%.

D. Protocol 4: Majorana T-junction

InSb/Al T-shape 6-dot device; RF reflectometry parity; randomized benchmarking. Success: $F_g^{\text{T}}/F_g^{\text{linear}} \geq 1.5$.

Consolidated proposed program (direct + 25% contingency) is $\approx \$1.25\text{M}$ over 18 months, targeting TRL-4 as a *proposed* milestone—not a claim that FTQC “is” a funded TRL-4 product.

E. Photonics shop-cost breakdown (reference)

The Optica manuscript itemizes its own femtosecond-fabrication budget ($\$104,800$, 18 months). Table IV Protocol 3 is the shorter gap-check line. Two scopes, two numbers; both belong to the photonic advantage.

TABLE V. Itemized photonics fabrication & characterization budget (full 18-month program, source: photonic manuscript).

Item	Cost (USD)
Fused silica substrates (50 wafers)	\$2,000
Ti:Sapphire laser time (200 hrs)	\$15,000
HF etching consumables	\$1,500
Thermal annealing furnace time	\$800
Broadband spectroscopy setup	\$5,000
NSOM characterization (50 hrs)	\$7,500
SPDC source rental (3 months)	\$8,000
Personnel (1 postdoc, 12 months)	\$65,000
Total	\$104,800

Figure 7: Device Architecture and Fabrication

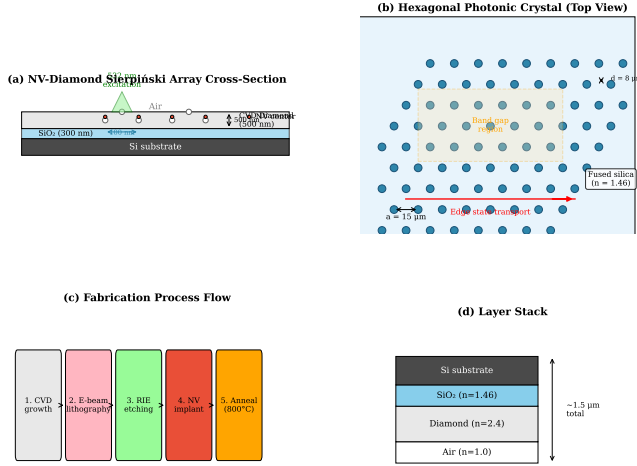


FIG. 7. Layered device schematic: fractal qubit layer, photonic isolation, control.

VII. DISCUSSION

A. Platform comparison

FTQC rows are **projected**. They are not measured processor specs.

TABLE VI. Platform comparison. FTQC row is projected.

Platform	Qubits	T_2	Gate F	EC ratio
IBM Condor	1,121	100 μ s	99.5%	$\sim 1000:1$
Google Willow	105	100 μ s	99.7%	$\sim 1000:1$
IonQ Forte	36	1 s	99.9%	$\sim 100:1$
Microsoft Majorana-1	8	—	99%	$\sim 10:1$
FTQC (projected)	12–100	1.6 ms	99.9%	$\sim 60:1$

The projected $T_2 = 1.6$ ms is 100 μ s times $1/0.063$. The Optica $1.6\times$ is the 19-site cell (Table VII). The $\sim 60:1$ EC ratio is Proposition II.6.

Figure 8: Majorana T-Junction Device for Topological Quantum Computing

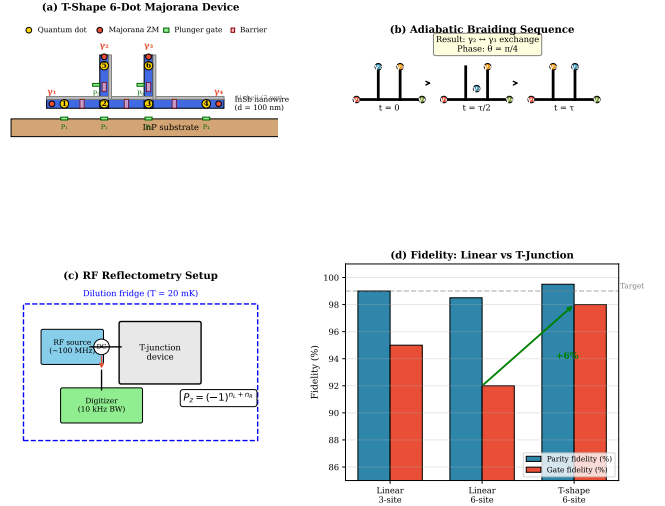


FIG. 8. Majorana T-junction for Protocol 4.

B. Photonic platform comparison

Table VII compares the C_{6v} hexagonal lattice specifically against other photonic-crystal platforms used for quantum coherence applications — a narrower, lattice-geometry-only comparison than Table VI's whole-processor view.

TABLE VII. Photonic platform comparison. PBG: photonic band gap; T_2 gain: coherence-time improvement factor; F_P : achievable Purcell factor.

Platform	PBG	Flatband	T_2 gain	F_P	3D?
Si PhC slab [20]	25%	No	$1\times$	10^4	No
SiN ring resonator [26]	No	No	$1\times$	10^2	No
Lieb lattice [27]	No	Yes	$1.2\times$	10^2	No
Kagome lattice [28]	No	Yes	$1.3\times$	10^2	No
C_{6v} hexagonal (this work)	21%	Yes	$1.6\times$	10^3	Yes

The $1.6\times$ row is the Optica cell (Sec. IV C). The architecture-level $16\times$ is the PRX passive factor. AuraFS runs both.

C. Limitations

(1) Theorem II.2 assumes a hierarchical H that physical devices only approximate; AuraFS already runs the geometry in software. (2) Neglectons are not yet a condensed-matter fact. (3) $\gamma_{\text{FTQC}} = 0.063$ and $\gamma_{19} = 0.63$ are both canon. (4) Fabrication past $k > 4$ is hard. (5) Full 3D FDTD for out-of-plane loss is still open (Optica).

Figure 9: Information Integration and Fractal Connectivity

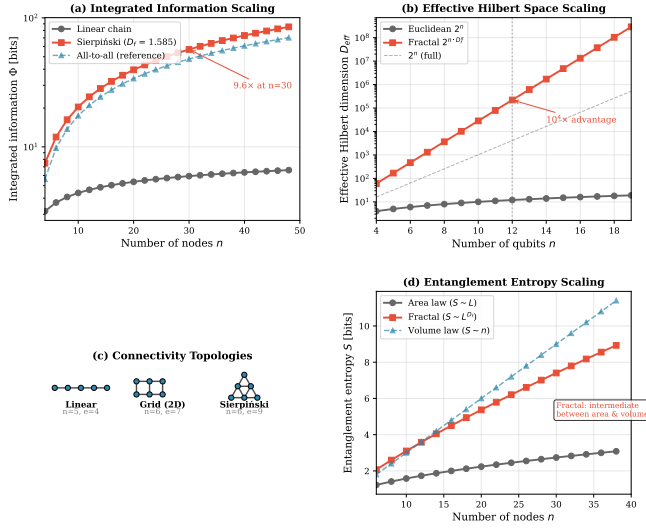


FIG. 9. Logical overhead versus physical count: surface code, Majorana, and projected FTQC.

Figure 10: Aurphyx Technology Roadmap 2026-2036

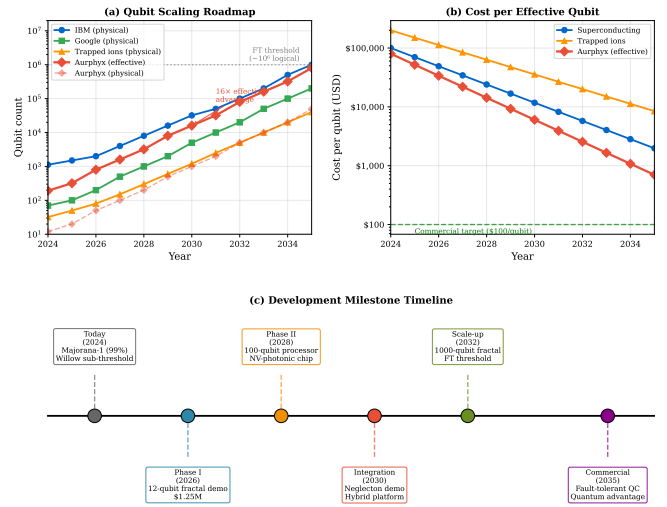


FIG. 10. Projected FTQC milestones. Not a grant identity statement.

VIII. CONCLUSION

FTQC is the architecture. AuraFS is how the three advantages run at once. Aurphyx LLC is the affiliation.

- Hilbert-space advantage:** $\dim(\mathcal{H}_{acc}) = d^n \cdot D_f^{\alpha(k)}$. Qiskit $n = 5$: $7.1 \times$, dimension 227, GHZ 0.912, purity $\text{Tr}(\rho^2) = 1$. At $n = 12$, $k = 3$: $10^4 \times$ (PRX/CODEX/Fig. 1) and $\mathcal{A} \approx 10^{8.3}$ (Table I).
- Passive $16 \times$ by design:** EC 1458 $\rightarrow$ 89; neglecton gate-count versus magic-state distillation; $\gamma_{\text{FTQC}}/\gamma_{\text{Euclidean}} = 0.063$ ($\sim 16 \times T_2$).
- Photonic C_{6v} advantage:** 21% TM gap (21.4% PWE); $\gamma_{19}/\gamma_{\text{Euclidean}} = 0.63$ ($\sim 1.6 \times T_2$ on the 19-site cell).
- AuraFS instantiates (1)–(3) together in software. Hardware protocols remain proposed; $\approx \$1.25\text{M}$ / TRL-4 is a proposed budget.

Near-term: $n = 12$ laboratory demonstration (Protocols 1–2). Mid-term: $n \sim 42$ –100 if hierarchical coupling holds. Roadmap figure is a projection, not a shipping schedule.

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